

**Syllabus**

Rational Numbers, Playing with Numbers, Squares and Square Roots, Cubes and Cube Roots, Exponents and Powers, Algebraic Expressions, Identities, Factorisation, Linear Equations in One Variable, Direct and Inverse Proportion, Commercial Mathematics, Visualising Solid Shapes, Mensuration.

**Instructions:**

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

**Mathematics**

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| <ol style="list-style-type: none"> <li>1. A rope of length <math>\frac{7}{8}</math> m is cut into 7 equal parts. What is the length of each part?<br/>             (A) <math>\frac{1}{8}</math> m                      (B) <math>\frac{7}{8}</math> m<br/>             (C) <math>\frac{1}{7}</math> m                      (D) <math>\frac{1}{64}</math> m</li> <li>2. A computer password system repeats a digit 4 times to generate a code. Which exponential form correctly represents this situation?<br/>             (A) <math>4^4</math>                      (B) <math>4 \times 4</math><br/>             (C) <math>4^3</math>                      (D) <math>4^2</math></li> <li>3. <b>Statement I:</b> The area of a triangle with base 10 cm and height 6 cm is 20 sq.cm.<br/> <b>Statement II:</b> Area of a triangle is equal to base <math>\times</math> height.<br/>             (A) Both Statement I and Statement II are true<br/>             (B) Both Statement I and Statement II are false<br/>             (C) Statement I is true and Statement II is false</li> </ol> | <ol style="list-style-type: none"> <li>(D) Statement I is false and Statement II is true</li> <li>4. A shopkeeper checks whether 2468 can be packed equally in 4 boxes. Which rule helps him decide quickly?<br/>             (A) Divisibility by 2<br/>             (B) Divisibility by 4<br/>             (C) Divisibility by 8<br/>             (D) Divisibility by 3</li> <li>5. A shopkeeper buys <math>4x + 8</math> notebooks and packs them equally in <math>x + 2</math> packets. How many notebooks are there in each packet?<br/>             (A) 2                      (B) 4<br/>             (C) 8                      (D) x</li> <li>6. Five times a number is equal to 45. What is the number?<br/>             (A) 5                      (B) 7<br/>             (C) 9                      (D) 15</li> </ol> |
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7. The area of a square garden is  $81 \text{ m}^2$ . What is the length of one side of the garden?  
 (A) 8 m (B) 9 m  
 (C) 10 m (D) 18 m
8. The A shopkeeper gives 20% discount on a shirt marked at ₹500. What is the selling price of the shirt?  
 (A) ₹400 (B) ₹450  
 (C) ₹480 (D) ₹420
9. Which of the following solids has only one curved surface and no flat surface?  
 (A) Cylinder (B) Cone  
 (C) Sphere (D) Hemisphere
10. A pack contains 48 chocolates. If they are distributed equally among 6 children, which concept is used?  
 (A) Prime factorisation  
 (B) Divisibility rule  
 (C) LCM  
 (D) HCF
11. A solid has 5 faces, 9 edges and 6 vertices. Identify the solid.  
 (A) Cube  
 (B) Triangular prism  
 (C) Square pyramid  
 (D) Tetrahedron
12. Riya has ₹120 more than twice the amount Aman has. If Aman has ₹x, which equation represents the situation?  
 (A)  $2x - 120$  (B)  $2x + 120$   
 (C)  $x + 120 = 2$  (D)  $120x = 2$
13. The librarian has 245 books. To arrange them in a perfect square, how many minimum books must be added?  
 (A) 5 (B) 11  
 (C) 15 (D) 21
14. If 4 workers can complete a job in 10 days, how many days will 2 workers take to complete the same job?  
 (A) 5 days (B) 10 days  
 (C) 15 days (D) 20 days
15. The distance between two planets is 420,000,000 km. Which is the correct standard form?  
 (A)  $4.2 \times 10^8 \text{ km}$   
 (B)  $42 \times 10^7 \text{ km}$   
 (C)  $0.42 \times 10^9 \text{ km}$   
 (D)  $420 \times 10^6 \text{ km}$
16. Five years ago, a person's age was half of his present age. What is his present age?  
 (A) 8 years (B) 10 years  
 (C) 12 years (D) 14 years
17. A school has 420 students and wants to form the largest possible equal groups. The number of students in each group should be the:  
 (A) LCM of 420  
 (B) HCF of factors of 420  
 (C) Prime factor of 420  
 (D) Multiple of 420
18. A classroom floor is 8 m long and 5 m wide. What is the area of the floor?  
 (A) 13 sq.m (B) 40 sq.m  
 (C) 26 sq.m (D) 80 sq.m

**19.** The total cost of buying equal number of pens and pencils is given by  $6x^2 - 15x$ . How many items are bought if the common factor is taken out?

- (A)  $3x(2x - 5)$  (B)  $x(6x - 15)$   
(C)  $15x$  (D)  $6(x - 5)$

**20.** The volume of a cube-shaped water tank is  $512 \text{ m}^3$ . What is the length of its side?

- (A) 6 m (B) 7 m  
(C) 8 m (D) 9 m

**21.** The price of a bag increases from ₹800 to ₹920. What is the percentage increase?

- (A) 12% (B) 15%  
(C) 18% (D) 20%

**22.** A bus travels  $5x$  km in one hour. It travels at the same speed for the first 4 hours. For the next 2 hours, its speed increases by  $x$  km per hour.

What is the total distance travelled by the bus in 6 hours?

- (A)  $30x$  (B)  $32x$   
(C)  $34x$  (D)  $36x$

**23.** A bus travels at a speed of  $x$  km/h for 4 hours and covers 200 km. Which equation represents this situation?

- (A)  $x + 4 = 200$   
(B)  $4x = 200$   
(C)  $x - 4 = 200$   
(D)  $x/4 = 200$

**24.** The perimeter of a rectangle is 48 cm. If its length is 14 cm, what is its breadth?

- (A) 8 cm (B) 10 cm  
(C) 12 cm (D) 14 cm

**25.** A rectangular garden has length 24 m and breadth 36 m. The longest possible square tiles used without cutting will have side:

- (A) 6 m (B) 12 m  
(C) 18 m (D) 24 m

**26.** In a science lab, energy is calculated as  $2^3 \times 2^5$ . Using laws of exponents, what is the simplified result?

- (A)  $2^8$  (B)  $2^{15}$   
(C)  $4^8$  (D) 16

**27.** A toy is made by placing a cone on top of a cylinder. How many curved surfaces does the toy have in total?

- (A) 1 (B) 2  
(C) 3 (D) 4

**28.** A water tank is filled by a pipe at the rate of  $\frac{5}{12}$  tank per hour. How much part of the tank will be filled in  $\frac{3}{4}$  hour?

- (A)  $\frac{5}{9}$  (B)  $\frac{15}{48}$   
(C)  $\frac{5}{16}$  (D)  $\frac{15}{12}$

**29.** Match the solid with the number of faces it has.

| Column I<br>(Solid) | Column II<br>(Number of Faces) |
|---------------------|--------------------------------|
| a) Cube             | 1) 1 curved face               |
| b) Cuboid           | 2) 6 faces                     |
| c) Sphere           | 3) 0 flat faces                |
| d) Cylinder         | 4) 3 faces                     |

- (A) a-2, b-2, c-3, d-4  
(B) a-2, b-4, c-3, d-1  
(C) a-4, b-1, c-3, d-1  
(D) a-2, b-1, c-4, d-3

- 30.** Find the Riya buys 3 notebooks costing ₹  $x$  each and a pen costing ₹10. Which algebraic expression represents the total cost?  
 (A)  $3 + x + 10$  (B)  $3x$   
 (C)  $3x + 10$  (D)  $x + 13$
- 31.** A square garden of side 6 m is attached to a rectangular garden of length 10 m and breadth 6 m. What is the total area of the two gardens?  
 (A) 96 sq.m (B) 120 sq.m  
 (C) 72 sq.m (D) 156 sq.m
- 32.** The length of a rectangle is  $(x+4)$  cm and the breadth is  $x$  cm. Which expression represents its perimeter?  
 (A)  $2x + 4$   
 (B)  $2(2x + 4)$   
 (C)  $2(x + 4)$   
 (D)  $4x + 4$
- 33.** A shopkeeper buys 1 kg sugar for Rs.  $\frac{48}{5}$ . What is the cost of  $\frac{3}{4}$  kg sugar?  
 (A) Rs.  $\frac{36}{5}$  (B) Rs.  $\frac{48}{5}$   
 (C) Rs.  $\frac{143}{20}$  (D) Rs.  $\frac{72}{5}$
- 34.** 12 workers can build a wall in 15 days. How many days will 20 workers take to build the same wall?  
 (A) 6 days (B) 8 days  
 (C) 9 days (D) 10 days
- 35.** A number is both a perfect square and a perfect cube. Which of the following could be the smallest such number?  
 (A) 16 (B) 36  
 (C) 64 (D) 81
- 36.** The parallel sides of a trapezium are 10 cm and 14 cm and the distance between them is 6 cm. Find its area.  
 (A) 72 sq.cm (B) 144 sq.cm  
 (C) 42 sq.cm (D) 84 sq.cm
- 37.** The volume of a box is calculated as  $(2 \times 5)^3$ . What is the correct simplified form?  
 (A)  $2^3 \times 5$  (B)  $2 \times 5^3$   
 (C)  $2^3 \times 5^3$  (D)  $10^3 \times 3$
- 38.** A jacket gets two successive discounts of 10% and 20% on a marked price of ₹1000. What is the final selling price?  
 (A) ₹700 (B) ₹720  
 (C) ₹750 (D) ₹800
- 39.** A cyclist covers  $\frac{15}{4}$  km in  $\frac{3}{2}$  hours. What is his speed in km/hour?  
 (A)  $\frac{5}{2}$  km/h (B) 5 km/h  
 (C)  $\frac{15}{6}$  km/h (D)  $\frac{10}{3}$  km/h
- 40.** Meena borrowed ₹4000 at 10% per annum for 3 years. How much interest will she pay?  
 (A) ₹1000 (B) ₹1100  
 (C) ₹1200 (D) ₹1300
- 41.** The area of a square plate is represented as  $(3^2)^3$ . What is the simplified exponential form?  
 (A)  $3^5$  (B)  $3^6$   
 (C)  $6^3$  (D)  $9^3$
- 42.** A farmer has 792 plants. He wants to plant them in a square field. What is the least number of plants he must remove to do so?  
 (A) 8 (B) 16  
 (C) 23 (D) 31

- 43.** A train takes 4 hours to cover a certain distance at a speed of 50 km/h. By how much should the speed be increased to cover the same distance in 2.5 hours?  
 (A) 20 km/h (B) 25 km/h  
 (C) 30 km/h (D) 35 km/h
- 44.** The area of a rectangular garden is given by the expression  $6x + 12$  square metres. What is the breadth of the garden if the length is  $x + 2$  metres?  
 (A) 6 (B) 2  
 (C) 3 (D) 12
- 45.** Three traffic lights change after 30 s, 45 s, and 60 s respectively. If they change together at 8:00 AM, at what time will they change together again?  
 (A) 8:03 AM (B) 8:05 AM  
 (C) 8:06 AM (D) 8:09 AM
- 46.** A bus takes 6 hours to travel a distance at a speed of 40 km/h. How long will it take to travel the same distance at 60 km/h?  
 (A) 3 hours (B) 4 hours  
 (C) 5 hours (D) 6 hours
- 47.** The cost of one apple is ₹ $x$  and one banana costs ₹ $y$ . If Ravi buys 5 apples and 3 bananas, which expression represents the total cost?  
 (A)  $8xy$  (B)  $5x + 3y$   
 (C)  $(x+y)^2$  (D)  $15xy$
- 48.** A shopkeeper buys an article for ₹600 and spends ₹50 on repairs. He sells it for ₹770. What is his profit percentage?  
 (A) 15% (B) 18%  
 (C) 20% (D) 25%
- 49.** The sum of two rational numbers is  $\frac{7}{6}$ . If one number is  $-\frac{1}{3}$ , find the other number.  
 (A)  $\frac{5}{6}$  (B)  $\frac{3}{2}$   
 (C)  $\frac{9}{6}$  (D)  $\frac{11}{6}$
- 50.** 8 men working 6 hours a day can complete a work in 10 days. How many days will 12 men working 4 hours a day take to complete the same work?  
 (A) 8 days (B) 9 days  
 (C) 10 days (D) 12 days

### KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) A  | 2) A  | 3) B  | 4) B  | 5) B  | 6) C  | 7) B  |
| 8) A  | 9) B  | 10) B | 11) B | 12) B | 13) B | 14) D |
| 15) A | 16) B | 17) B | 18) B | 19) A | 20) C | 21) B |
| 22) B | 23) B | 24) B | 25) B | 26) A | 27) B | 28) C |
| 29) A | 30) C | 31) A | 32) B | 33) A | 34) C | 35) C |
| 36) A | 37) C | 38) B | 39) A | 40) C | 41) B | 42) A |
| 43) C | 44) A | 45) A | 46) B | 47) B | 48) B | 49) C |
| 50) C |       |       |       |       |       |       |